
Coord-EfficientNet-B3: Proposed DeepFake Detection Architecture
Input Facial Image
RGB Face Crop
224 × 224 × 3
EfficientNet-B3 Backbone
ImageNet Pre-trained
Stem + MBConv Stages 1–7
Output Feature Map:
F ∈ ℝ^1536 × 7 × 7
Proposed Coordinate Attention (CoordAtt) Module
X-AvgPool (1D)
Pool Width W → 1
z^h ∈ ℝ^1536 × 7 × 1
Y-AvgPool (1D)
Pool Height H → 1
z^w ∈ ℝ^1536 × 1 × 7
Concat & Shared 1×1 Conv
Batch Normalization
Hard-Swish Activation
Reduction Ratio r = 32
1536 → 48 channels
Horizontal 1×1 Conv
Sigmoid Activation
48 → 1536 channels
g^h ∈ ℝ^1536 × 7 × 1
Vertical 1×1 Conv
Sigmoid Activation
48 → 1536 channels
g^w ∈ ℝ^1536 × 1 × 7
Identity Feature Map F
Feature Modulation
⊗
Y = F ⊗ g^h ⊗ g^w
Y ∈ ℝ^1536 × 7 × 7
Global Average Pooling (GAP)
ỹ ∈ ℝ^1536
3-Stage Classifier Head
Dense 1536 → 128
ReLU + Dropout (0.3)
Dense 128 → 64
ReLU + Dropout (0.2)
Dense 64 → 1 (Logit z)
Classification Decision
P(Fake | X) = σ(z)
Threshold τ = 0.50
τ < 0.50 → Real
τ ≥ 0.50 → Fake